

CLEANING DIRTY WATER

WASTE WATER TREATMENT

Billions of gallons of dirty water from industries, farms, and homes flow into drains and sewers every day. This water is a mixture of human sewage and all kinds of dangerous bacteria and chemicals, so it must be thoroughly cleaned before it can be released back into the environment. Treating waste water is essential to prevent water scarcity. The multistage process uses several physical, biological, and chemical methods to decontaminate the water and make it safe for reuse. One stage separates the solid waste from the water using a process called sedimentation.

SLUDGE CAKE

The dry solid waste left over from water treatment is called sludge cake and has its own use. Sludge cakes can be used as fertilizers to help plants grow by providing them with extra nutrients.



SEDIMENTATION

Sedimentation is a physical method of cleaning water in which heavy solid particles in a solution sink to the bottom, pulled by gravity, so they can be removed easily. When dirty water passes into a sedimentation tank (below), a sludge of solid particles, such as those from human sewage, settles at the bottom to be collected, while lighter substances such as oil and scum float to the top to be skimmed off.

